Activity-4

Binary tree & AVL Tree

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1) AVL Tree

keys: 70, 50, 30, 90, 80, 130, 120 & deletion of 120 & 30.

steps:

step 1: 70

step 2: Insert 50

```
    70
   /
 50
```

step 3: Insert 30

```
      70
     /
   50      RR       50
  /        =>      /  \
30               30    70
```

step 4: Insert 90

```
    50
   /  \
 30    70
         \
          90
```

step 5: Insert 80

```
  50
 /  \
30   70      RL
      \
       90
      /
    80
```

RR at 90

```
  50
 /  \
30   70
      \
       80
        \
         90
```

LR at 70

```
      50
     /  \
   30    80
        /  \
      70    90
```

step 6 Insert 130

```
    50
   /  \
 30    80
      /  \
    70    90
            \
             130
```

step 7 Insert 120

```
   50                       50
  /  \        LR 90        /  \
30    80       =>        30    80
     /  \                     /  \
   70    90                 70    130
           \                     /
            130                90
           /                     \
        120                       120
```

step 8 Delete 120

```
    50
   /  \
 30    80
      /  \
    70    130
          /
        90
```

step 9: Delete 30

```
  50                    50
    \                     \
     80                    80
    /  \                  /  \
  70    130             70    130
        /                     /
      90                    90
```

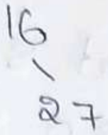
2) AVL Tree

keys: 16, 27, 9, 11, 36, 54, 81, 63, 72.

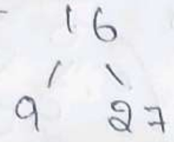
Step 1:

16

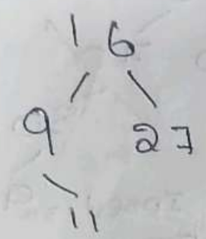
Step 2:



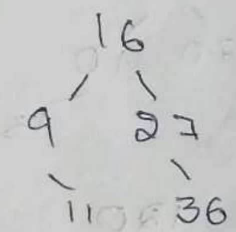
Step 3:



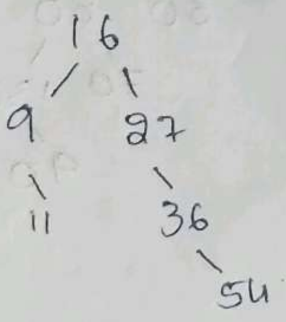
Step 4: Insert 11

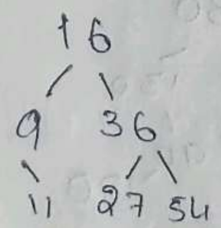


Step 5: Insert 36



Step 6: Insert 54

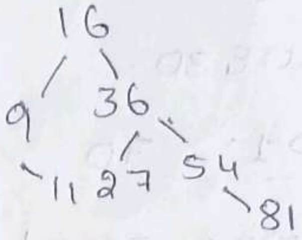




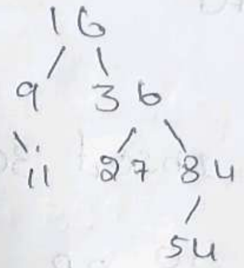
At 27 → RR

L Rotation at 27

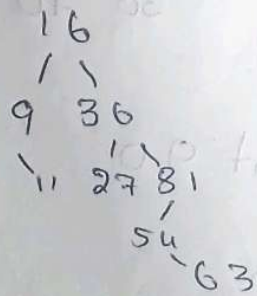
Step 7: Insert 81

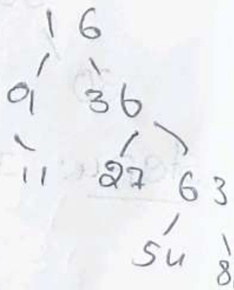


Left rotation at 54

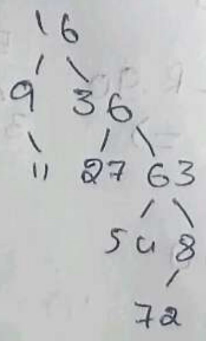


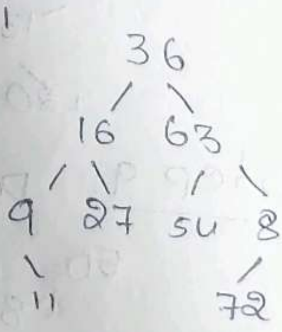
Step 8: Insert 63





Step 9: Insert 72





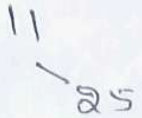
3) Binary search tree

keys: 11, 25, 10, 15, 39, 58, 84, 65, 78

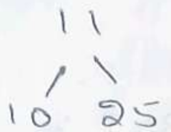
step1 Insert 11

11

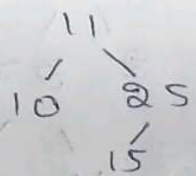
step2: Insert 25



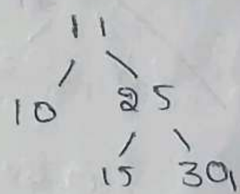
step3: Insert 10



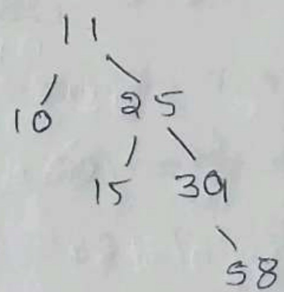
step4: Insert 15



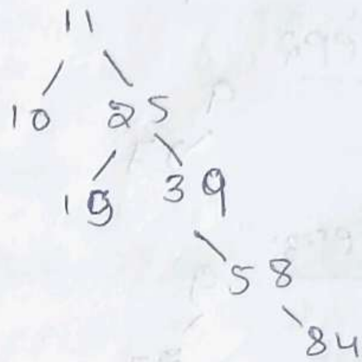
step5: Insert 39



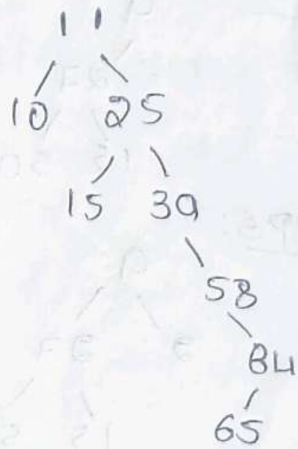
step6: Insert 58



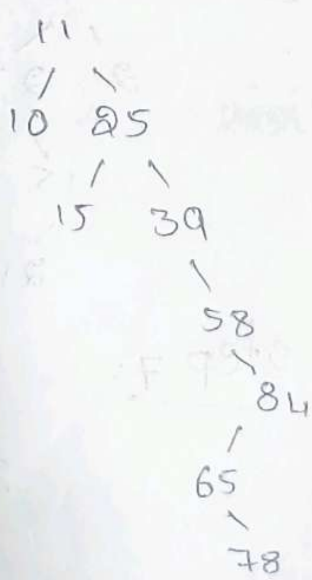
step7: Insert 84



step 8: Insert 65



step 9: Insert 78

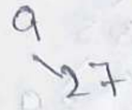


4) Binary search tree

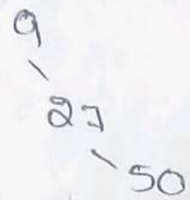
Key: 9, 27, 50, 15, 2, 21, 36

step 1:

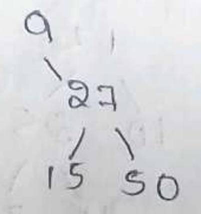
step 2



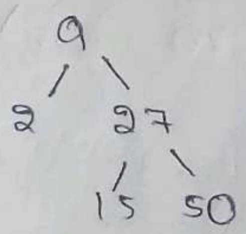
step 3:



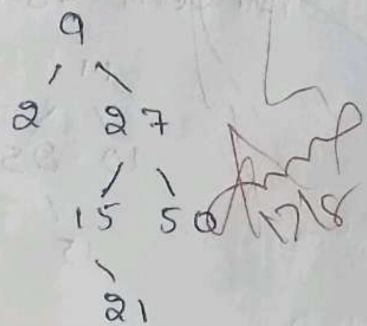
step 4:



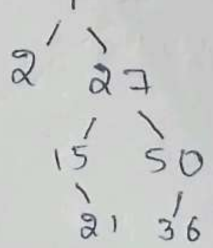
step 5:



step 6



step 7:



Final BST: